Christopher Carney ([illegible])

DC27

Statistics & Data Analysis

Assignment 1

Grade: REVIEW_REQUIRED

Review Notes:

x Qa: Review manually.

x Qb: Review manually.

x Qc: Review manually.

x Qd: Review manually.

x Qe: Review manually.

a. $P(60)+P(80)+P(100) = 0.15+0.25+0.05 = \boxed{0.45}$

b. $\mu = \sum x f(x) = (10\cdot 0.10)+(20\cdot 0.15)+(40\cdot 0.30)+(60\cdot 0.15)$
$+(80\cdot 0.25)+(100\cdot 0.05) \boxed{= 50}$

$\sigma^2 = \sum (x-\mu)^2 f(x) = (10-50)^2(0.10)+(20-50)^2(0.15)+(40-50)^2(0.30)$
$+(60-50)^2(0.15)+(80-50)^2(0.25)+(100-50)^2(0.05)$
$= 160+135+30+15+225+125$

$\sigma^2 = 690$

$\sigma = \sqrt{690} \boxed{= 26.27}$

c. Expected value is the mean, your long term weighted average of the amount withdrawn. It is <u>not</u> a value any single customer actually withdraws.

d. $P(50-26.27 \le \$ \le 50+26.27) = P(23.73 \le \$ \le 76.27)$
$= P(40)+P(60)$
$= \boxed{0.45 \text{ or } 45\%}$

e. $P(50-2\cdot 26.27 \le \$ \le 50+2\cdot 26.27) = P(-2.54 \le \$ \le 102.54)$
$= P(10)+P(20)+P(40)+P(60)+P(80)+P(100)$
$= \boxed{1.0 \text{ or } 100\%}$